Assignment–7

Design of OPAMP for Mathematical Operation

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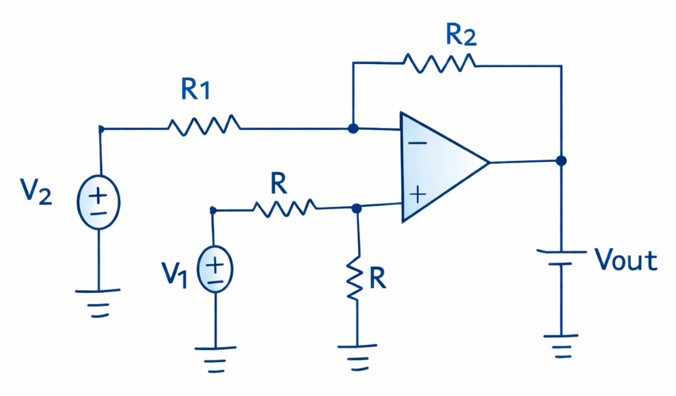
Aim

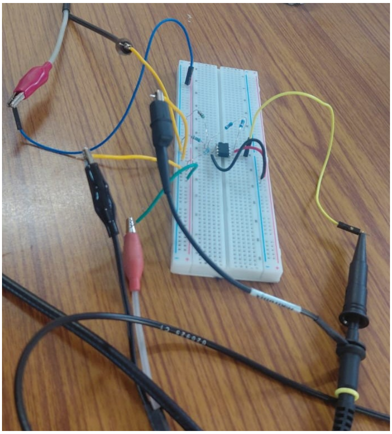
To design an operational amplifier circuit that performs the mathematical operation

$$V_o = 2V_1 - 3V_2$$

and test it using a DC source and a low frequency sinusoidal signal.

Circuit





Derivation

From the input network,

$$V_1 - iR - iR = 0 \Rightarrow i = \frac{V_1}{2R}$$

Voltage at non-inverting terminal:

$$V_+ = V_1 - iR = V_1 - \frac{V_1}{2} = \frac{V_1}{2}$$

Using virtual short:

$$V_- = V_+ = \frac{V_1}{2}$$

Applying KCL at inverting node,

$$\frac{V_2 - \frac{V_1}{2}}{R_1} = \frac{\frac{V_1}{2} - V_{out}}{R_2}$$

$$\Rightarrow V_{out} = \frac{R_1 + R_2}{2R_1} V_1 - \frac{R_2}{R_1} V_2$$

For required operation $V_{out} = 2V_1 - 3V_2$,

$$\frac{R_1 + R_2}{2R_1} = 2, \qquad \frac{R_2}{R_1} = 3$$

$$\Rightarrow R_2 = 3R_1$$

$$\boxed{\frac{R_2}{R_1} = 3}$$

To satisfy this condition, the following practical resistor values were used in the circuit:

- $R_1 = 10\,k\Omega$
- $R_2 = 30\,k\Omega$
- $R = 1\,k\Omega$

Since a single $30\,k\Omega$ resistor was not available, it was implemented by connecting two $15\,k\Omega$ resistors in series

Testing Using DC Source

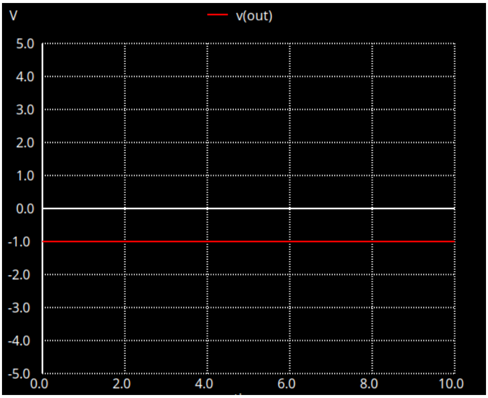
To verify the designed Op-Amp circuit for the operation

$$V_o = 2V_1 - 3V_2$$

DC voltages were applied as inputs V_1 and V_2. The output voltage V_o was measured using an oscilloscope. Different combinations of input voltages were applied to verify the required operation.
For the applied DC input values:

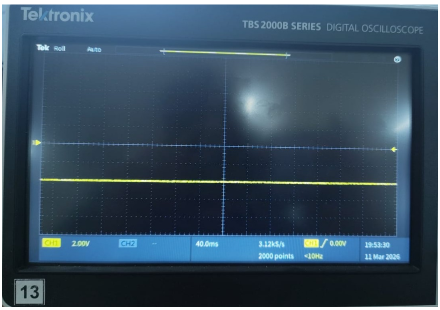
1. When $V_1 = 1\,V$ and $V_2 = 1\,V$

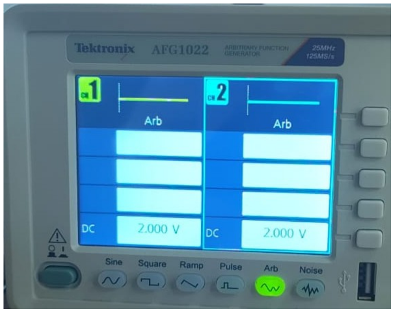
$$V_o = 2(1) - 3(1) = -1\,V$$

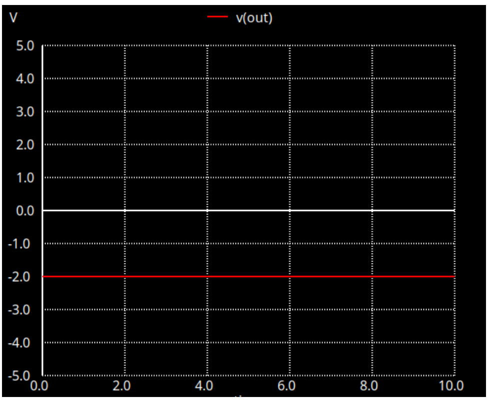


2. When $V_1 = 2\ V$ and $V_2 = 2\ V$

$$V_o = 2(2) - 3(2) = -2\ V$$

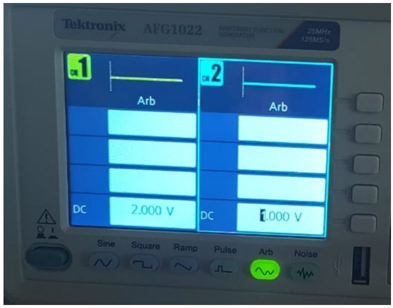


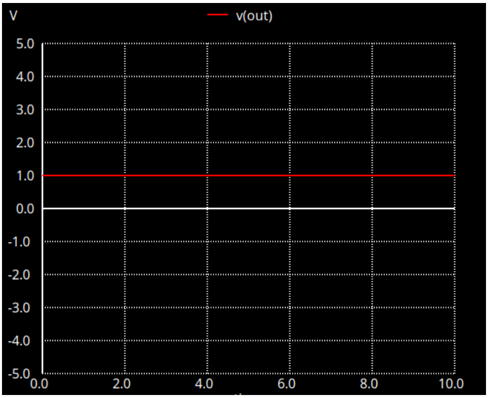




3. When $V_1 = 2\,V$ and $V_2 = 1\,V$

$$V_o = 2(2) - 3(1) = 1\,V$$





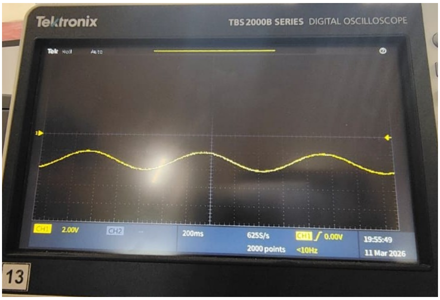
The measured output voltages matched the theoretical values, confirming that the Op-Amp circuit successfully performs the required operation.

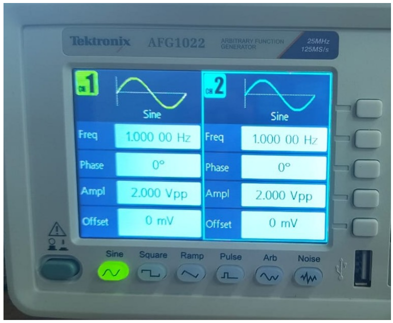
Testing Using Low Frequency Sinusoidal Signal

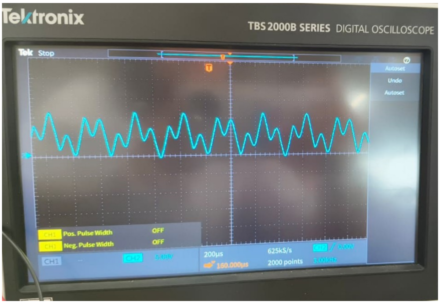
The Op-Amp circuit was tested using low frequency sinusoidal signals applied to V_1 and V_2 using a function generator. The output was observed on a digital oscilloscope. A single output waveform was observed, which follows the relation

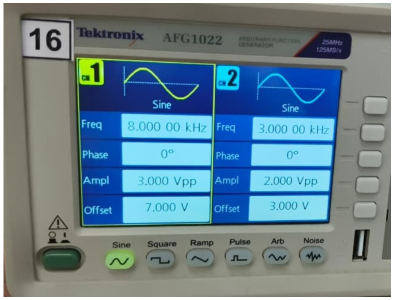
$$V_o = 2V_1 - 3V_2$$

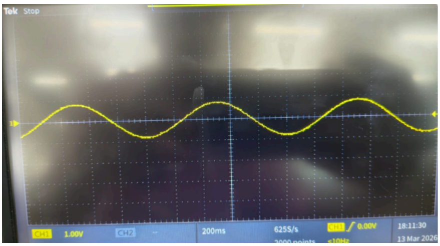
This verifies that the circuit performs the required operation correctly.

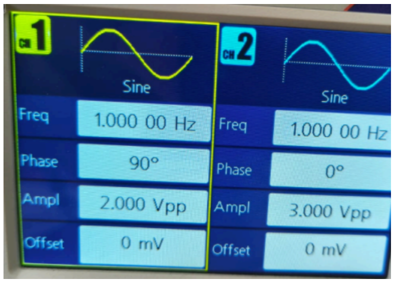

Tektronix
TBS 2000B SERIES DIGITAL OSCILLOSCOPE
13
CH1 2.00V
CH2
200ms
625S/s
2000 points
CH1 0.00V
<10Hz
19:55:49
11 Mar 2026


Tektronix AFG1022
CH 1
Sine
Freq 1.000 00 Hz
Phase 0°
Ampl 2.000 Vpp
Offset 0 mV
CH 2
Sine
Freq 1.000 00 Hz
Phase 0°
Ampl 2.000 Vpp
Offset 0 mV
Sine Square Ramp Pulse Arb Noise


Tektronix
TBS 2000B SERIES DIGITAL OSCILLOSCOPE
Tek Stop
Autoset
Undo
Autoset
CH1 Pos. Pulse Width OFF
CH1 Neg. Pulse Width OFF
CH1
CH2
200µs
625kS/s
2000 points


16
Tektronix AFG1022
CH 1
Sine
Freq 8.000 00 kHz
Phase 0°
Ampl 3.000 Vpp
Offset 7.000 V
CH 2
Sine
Freq 3.000 00 kHz
Phase 0°
Ampl 2.000 Vpp
Offset 3.000 V
Sine Square Ramp Pulse Arb Noise


Tek Stop
CH1 1.00V
CH2
200ms
625S/s
CH1 0.00V
18:11:30
13 Mar 2026


CH 1
Sine
Freq 1.000 00 Hz
Phase 90°
Ampl 2.000 Vpp
Offset 0 mV
CH 2
Sine
Freq 1.000 00 Hz
Phase 0°
Ampl 3.000 Vpp
Offset 0 mV